

Computational Verification Note

Fully Expanded Standard Model Lagrangian with Three Right-Handed Dirac Neutrinos

Result: **PASS**

All 24 parts of the submitted Lagrangian agreed exactly with the independently constructed reference calculation, allowing for integration by parts.

Purpose

This note records the computational verification of the accompanying fully expanded Lagrangian. The model is the Standard Model extended by three right-handed neutrino fields. The new fields are singlets under the Standard Model gauge groups, and neutrino masses are Dirac masses produced by Yukawa interactions. Majorana mass terms are not part of the expression that was checked.

The purpose of the calculation was to determine whether the long expanded expression has the terms, coefficients, signs, indices, chiral structure and flavour-matrix ordering required by its stated model and conventions.

How the calculation was checked

The submitted source expression was not used as the formula from which the reference answer was generated. A separate symbolic calculation began with the declared gauge groups, fields, representations, interactions, symmetry breaking, field rotations, gauge-fixing functions and flavour rotations. These declarations used the same published conventions as the submitted document so that the two expressions represented the same mathematical theory. The expanded submitted expression was then compared with the result of that calculation.

The comparison was divided into 24 physically distinct parts. They cover the QCD and electroweak gauge terms; scalar kinetic, potential and gauge interactions; gauge fixing; QCD and electroweak ghost terms; fermion kinetic, mass and gauge-current terms; and Higgs and Goldstone Yukawa interactions. Each part had to agree independently, so an error in one part could not be hidden by an opposite error elsewhere.

The calculation used exact symbolic coefficients rather than decimal approximations. Declared identities—including the weak-angle relation and the relations between the mass, vacuum and Higgs-potential parameters—were fixed before comparison. Terms related by dummy-index renaming and declared tensor symmetries were identified. Integration by parts was allowed only when its discarded total derivative was recorded.

Result

For every one of the 24 parts, the submitted expression minus the independently calculated expression reduced exactly to a total derivative:

$$\mathcal{L}_{\text{submitted}} - \mathcal{L}_{\text{calculated}} = \partial_\mu K^\mu, \quad R_s = 0 \quad \text{in every part } s.$$

Here R_s is the non-boundary remainder in part s . All of these remainders were exactly zero. Twenty-six nonzero boundary contributions were retained to document the allowed integration-by-parts rearrangements.

Summary of the evidence

Check	Recorded result
Submitted expression successfully imported	Pass
Separate reference calculation completed	Pass
Independent parts compared	24 of 24
Non-boundary remainder in every part	Exactly zero
Overall non-boundary remainder	Exactly zero
Recorded integration-by-parts contributions	26
Hermiticity in the stated scope	Pass
Deliberately incorrect test cases detected	1,000 of 1,000
Equivalent rearrangements accepted	120 of 120
Mathematically invalid test cases rejected	80 of 80
Separate check of the saved calculation record	Pass

The separate check of the saved calculation record did not call the main routines that generated or simplified the reference expression. It checked the recorded mathematical steps and final comparisons using a restricted checker. This provides an additional check against corruption or an unsupported result, although it is not a second independently developed derivation of the model.

What the result establishes

Within the field content, conventions, identities and assumptions stated in the submitted document, the expanded Lagrangian is mathematically consistent with the separately constructed Standard Model plus three right-handed Dirac-neutrino calculation. The check specifically provides evidence against omitted or additional terms, incorrect numerical or symbolic factors, sign mistakes, invalid index contractions, wrong chiral projectors, incorrect mixing-matrix order, and inconsistent gauge-fixing or ghost interactions.

Identification of the verified expression

The result applies only to the exact source file identified below. Any change to that file requires a new verification.

Source file	SM_RHNu_Dirac_v1.tex
Source SHA-256	a585bb6e3c9b5c49fa2eae6d706ddfc9e9ee3b73d2c56f32c6a7e484865b54e
Compared-expression SHA-256	66f5d6ea12560d76d173a730e446ea9f5429e4c1516fb3aa083dd545c5f2e69d
Verification date	4 September 2026
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